Measuring productivity

Lindeman's trophic-dynamic theory helped to clarify the idea of ecosystem productivity, which ecologists had previously defined in rather vague terms. The productivity of a plant or animal is measured by its growth in organic material, or biomass. This is never equal to the organism's energy input, because the conversion of solar energy into leaf in the case of plants, or the conversion of food into flesh in the case of an animal, is never 100 percent efficient. Some energy is released as heat, most of which is lost via respiration—an essential aspect of metabolism in all living things.

Warm-blooded animals lose a lot of heat when their body temperature is much higher than that of their surroundings. All animals also lose energy when they excrete urine. In addition, not all the material in an animal's food can be digested in its gut, and the material that is expelled as feces represents unused chemical energy.